

DSA-210 Final Report

Gender Bias in Online Toxicity Analysis

1. Motivation

Online communication platforms have become one of the most important spaces for social interaction. However, online environments also contain toxic language, harassment, and hate speech. In recent years, discussions about gender bias and online misogyny have become increasingly common on social media platforms, forums, and comment sections. Because of this, I wanted to investigate whether comments containing female-related expressions tend to exhibit different toxicity patterns compared to general online comments. I selected this topic for two main reasons. First, it is socially meaningful because gender bias in online communication affects how individuals experience digital platforms. Second, it is also an interesting data science problem because it combines text processing, feature engineering, exploratory data analysis, statistical hypothesis testing, and machine learning. The project focuses on whether simple textual indicators related to women can reveal measurable differences in toxicity levels. Rather than building a highly optimized NLP system, the purpose of this study was to construct a complete data science pipeline and interpret the findings in a meaningful way.

2. Data Source

The project uses the publicly available Jigsaw Toxic Comment Classification Dataset from Kaggle. This dataset contains thousands of online comments labeled with different toxicity categories. The labels include: • toxic • severe_toxic • obscene • threat • insult • identity_hate The dataset is widely used in NLP and online toxicity research because it provides a large-scale collection of labeled comments suitable for both statistical analysis and machine learning classification tasks. The original dataset was downloaded locally and processed in Python. Due to file size limitations, the raw train.csv file was not uploaded to the GitHub repository. Instead, the repository contains the executed notebook, analysis code, visualizations, and report files.

3. Data Preparation and Feature Engineering

Several preprocessing and feature engineering steps were applied before the analysis stage. First, comment texts were converted to lowercase in order to standardize the textual format. Then, a keyword-based filtering approach was used to identify comments potentially related to women. The selected keywords included: • she • woman • women • female • girl Using these keywords, a new binary feature called women_related was created. If a comment contained at least one of the selected expressions, it was labeled as women-related. Afterward, a combined toxicity score was calculated by averaging the following toxicity-related labels: • toxic • severe_toxic • obscene • threat • insult • identity_hate This combined score was used throughout the project as a simplified numerical representation of overall toxicity.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to better understand the dataset structure and toxicity patterns. Several visualizations were created during this stage: • Comment group count comparison • Average toxicity score comparison • Toxicity score histogram • Toxicity score boxplot The histogram visualizations showed that most comments have relatively low toxicity scores, while a smaller number of comments contain extremely toxic content. This resulted in a skewed distribution with visible outliers. The boxplots revealed differences in variability between women-related comments and other comments. In addition, the average toxicity comparison suggested that there

may be measurable differences between the groups. EDA was an important stage because it provided initial evidence that the two groups might behave differently in terms of toxicity distribution.

5. Hypothesis Testing and Statistical Analysis

To move beyond visual inspection, several statistical tests were applied. Before applying parametric tests, normality assumptions were evaluated using the Shapiro-Wilk normality test. The results suggested that the toxicity score distributions were not perfectly normal. Because of this, both parametric and non-parametric tests were included in the analysis. The following tests were applied: • Independent t-test • Mann-Whitney U test The independent t-test was used to compare average toxicity scores between women-related comments and other comments. The results indicated statistically significant differences between the groups. The Mann-Whitney U test was also applied because it does not require strict normality assumptions. This test supported the findings of the t-test and confirmed that the distributions differed significantly. Using multiple statistical methods strengthened the reliability of the analysis.

6. Machine Learning Methods

After the EDA and hypothesis testing stages, machine learning methods were applied to classify toxic comments. The target variable was the toxic label from the dataset. The selected features included: • `women_related` • `toxicity_score` The dataset was split into training and testing sets using an 80/20 train-test split. Two machine learning models were implemented: • Logistic Regression • Decision Tree Classifier Logistic Regression served as a simple baseline classification model. Decision Tree was included as a nonlinear alternative capable of capturing more flexible relationships. The models were evaluated using: • Accuracy score • Classification report • Confusion matrix The purpose of this stage was not only to maximize prediction accuracy, but also to observe whether simple textual indicators and toxicity-related features can help classify toxic comments.

7. Results and Findings

The results suggest that women-related comments exhibit different toxicity distributions compared to general online comments. The statistical analysis showed that the differences between groups were statistically significant according to both the independent t-test and the Mann-Whitney U test. The machine learning models achieved reasonable performance in predicting toxic comments. Logistic Regression provided a simple and interpretable baseline, while the Decision Tree model showed greater flexibility in handling nonlinear relationships. Although the project does not claim causal conclusions about gender bias, the findings indicate that comments containing female-related expressions may exhibit distinct toxicity patterns within online communication datasets.

8. Discussion

One of the most important aspects of this project is that it combines both statistical analysis and machine learning methods within a single workflow. The EDA and hypothesis tests suggested that toxicity distributions differ between the selected groups. At the same time, the machine learning models demonstrated that these features contain predictive information. However, the project also revealed several limitations in feature design. The keyword-based filtering method is intentionally simple and may not fully capture context. Some comments may contain female-related words without directly referring to women, while others may imply gender-related content without using explicit keywords. In addition, toxicity itself is a complex and context-dependent concept. Simple numerical representations cannot fully capture sarcasm, irony, or conversational meaning. Despite these limitations, the project successfully demonstrates how data science methods can be applied to investigate social patterns within online communication.

9. Limitations and Future Work

This project has several limitations. First, the women-related feature was created using a small keyword list. More advanced NLP approaches could improve contextual understanding. Second, the combined toxicity score simplifies multiple toxicity categories into a single numerical variable. While this makes analysis easier, it may also remove category-specific nuances. Third, the machine learning models used in this project were intentionally simple. More advanced models could potentially achieve stronger predictive performance. Possible future improvements include:

- Using transformer-based NLP models such as BERT
- Expanding the keyword filtering system
- Applying TF-IDF and embedding-based approaches more extensively
- Including sentiment analysis and contextual embeddings
- Comparing results across multiple online platforms
- Performing hyperparameter optimization for machine learning models

10. Conclusion

This project investigated whether comments containing female-related expressions exhibit different toxicity patterns compared to other online comments. The workflow included data preprocessing, feature engineering, exploratory data analysis, statistical hypothesis testing, and machine learning classification. The findings suggest that women-related comments may differ in toxicity distribution and that these differences can be detected both statistically and computationally. More importantly, the project demonstrates how data science techniques can be applied to socially relevant questions involving online communication, bias, and toxic language.

11. AI Usage Disclosure

AI tools were used during the project for debugging support, improving report organization, refining wording, and assisting with technical explanations. However, dataset preparation, preprocessing decisions, notebook execution, statistical analysis, visualization generation, machine learning implementation, interpretation of findings, and final review of the project were completed and supervised by the author.

Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Processing
Matplotlib	Visualizations
SciPy	Hypothesis Testing
Scikit-learn	Machine Learning
Jupyter Notebook	Interactive Analysis